

Deep-Learning Diagnosis Across Diseases: A CNN–GNN Hybrid Benchmark and a Confident-Learning Label-Noise Census of Two Public Medical Image Datasets

MEGA-PROGRAM-27, Item 10 (tools 4–5): malaria parasite detection and pneumonia detection

MEGA27-10b build lane (computational study; no wet lab)

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Abstract

We build and verify two deep-learning diagnosis tools on fully public data: (10.4) malaria parasite detection on the NIH Lister Hill cell-image dataset (27,558 images) and (10.5) pneumonia detection on the Kermamy et al. 2018 chest X-ray dataset (5,856 images, official patient-level split, integrity verified by SHA-256 against the publisher-stated hash). Each tool pairs a compact convolutional trunk with a region-graph neural network head whose propagation rule we derive and analyse. Beyond reproducing published benchmarks, we perform a *label-noise census*: a confident-learning audit that names, by named identifier, individual images whose labels are statistically inconsistent with cross-validated model confidence, quantifies the noise rate of each dataset, and measures the retraining delta after correction. All code, splits, result files and figures are committed to the project repository; every number in this paper is generated from a committed JSON artefact. Negative results are preserved as findings.

Contents

1	Introduction	3
1.1	Scope and items 10.4–10.5	3
1.2	Claims audited	3
1.3	Contribution	3
2	Mathematical foundations	3
2.1	Convolutional feature extraction	3
2.2	Region-graph construction	3
2.3	GCN propagation and spectrum	4
2.4	Objective and optimisation	4
2.5	Confident learning for label noise	4
2.6	Evaluation metrics	5
3	Data	5
3.1	Malaria (item 10.4)	5
3.2	Pneumonia (item 10.5)	5
3.3	Provenance and integrity	5

4	Methods	5
4.1	Models	5
4.2	Training protocol	5
4.3	Census protocol	5
5	Results	5
6	Discussion	6
6.1	What the census is and is not	6
6.2	The admissibility gate	6
6.3	Engineering for a two-core sandbox	6
6.4	Uncertainty quantification	6
6.5	Interpretability check	7
6.6	Malaria census findings	7
6.7	Pneumonia: an honest gap, and where it lives	8
6.8	The hybrid’s value is dataset-dependent	8
6.9	Honest negatives	8
7	External tools and datasets	9
8	Reproducibility	10
A	Flagged identifiers	10

1 Introduction

1.1 Scope and items 10.4–10.5

This report covers tools 4 and 5 of item 10 of the MEGA-PROGRAM-27 computational-biology program: deep-learning diagnosis across diseases. Lane 10a covers tools 1–3 on tabular cohorts (WDBC, Cleveland, Pima) and a 22-dataset panel; the present lane deliberately selects *disjoint* diseases and data modalities — microscopy cell images (malaria) and radiographs (pneumonia) — so the two lanes together span tabular and image diagnosis without duplication.

1.2 Claims audited

For malaria, Rajaraman et al. (2018) report $\approx 95.9\%$ accuracy for a custom CNN and $\approx 97.4\%$ for a pre-trained VGG-19 on the NIH dataset. For pneumonia, Kermany et al. (2018) report 92.8% test accuracy (sensitivity 93.2%, specificity 90.1%) for a transfer Inception-v3 on the 624-image official test set. We treat these as the published frontier and report head-to-head numbers on the same official splits.

1.3 Contribution

1. A verified benchmark reproduction on integrity-checked data (SHA-verified downloads; splits committed).
2. A compact CNN + region-GNN hybrid ($\approx 159k$ parameters) whose propagation rule we derive; it runs on 2 CPU cores.
3. A **label-noise census** of both datasets: named, individually checkable image identifiers flagged by a from-scratch confident-learning estimator, with quantified noise rates and measured retraining deltas. To our knowledge no published census of these label errors exists.

2 Mathematical foundations

2.1 Convolutional feature extraction

Given an input image $x \in \mathbb{R}^{C_0 \times H_0 \times W_0}$, a convolutional layer computes

$$f_{c,i,j}^{(l)} = \phi \left(b_c^{(l)} + \sum_{c'=1}^{C_{l-1}} \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} W_{c,c',u,v}^{(l)} f_{c',i+u,j+v}^{(l-1)} \right), \quad (1)$$

with ϕ the ReLU nonlinearity, $k = 3$ kernels, and batch normalisation $\hat{f} = \gamma(f - \mu_B) / \sqrt{\sigma_B^2 + \varepsilon} + \beta$ between layers. Max pooling halves spatial resolution three times, so a 64×64 input yields an 8×8 feature map with $C=128$ channels.

2.2 Region-graph construction

Definition 1 (Region graph). Let $F \in \mathbb{R}^{C \times H \times W}$ be the CNN feature map and g the grid size. Node $v_{r,c}$ ($0 \leq r, c < g$) has embedding

$$h_{r,c}^{(0)} = W_p \text{avgpool}(F; r, c) \in \mathbb{R}^d, \quad \text{avgpool}(F; r, c) = \frac{g^2}{HW} \sum_{i \in R_r} \sum_{j \in R_c} F_{:,i,j}, \quad (2)$$

and edge set E joins all 8-neighbour grid pairs (king moves on the grid).

Proposition 1 (Adjacency moments). *For a $g \times g$ grid with 8-neighbour adjacency A : (i) A is symmetric with zero diagonal; (ii) corner nodes have degree 3, edge (non-corner) nodes degree 5, interior nodes degree 8; (iii) the number of edges is $|E| = 8g^2 - 12g + 4$ for $g \geq 2$.*

Proof. (i)–(ii) follow from the bijection $(r, c) \leftrightarrow (r', c')$ of king moves and boundary counting. For (iii), sum degrees: interior $(g-2)^2$ nodes contribute $8(g-2)^2$; the $4(g-2)$ edge nodes contribute $5 \cdot 4(g-2)$; corners contribute 12. Handshaking gives $|E| = \frac{1}{2}[8(g-2)^2 + 20(g-2) + 12] = 8g^2 - 12g + 4$. $\square$

2.3 GCN propagation and spectrum

Definition 2 (Normalised propagation). With $\tilde{A} = A + I$, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$,

$$\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A} H^{(l)} W^{(l)}). \quad (3)$$

Lemma 1 (Spectral bound). *The eigenvalues of $\hat{A}$ lie in $(-1, 1]$.*

Proof sketch. $\hat{A} = I - \tilde{D}^{-1/2} \tilde{L} \tilde{D}^{-1/2}$ where $\tilde{L} = \tilde{D} - \tilde{A}$ is positive semidefinite; the normalised Laplacian of a connected non-bipartite weighted graph with self loops has spectrum in $[0, 2)$, hence $\lambda(\hat{A}) \in (-1, 1]$. Self loops shrink the largest Laplacian eigenvalue below 2 (Kipf & Welling 2017, App. A). $\square$

Repeated application of (3) contracts high-frequency components, which is why two layers suffice at our graph scale ($g=4$, 16 nodes, $|E| = 100$ by Proposition 2.2).

2.4 Objective and optimisation

We minimise the cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_{n=1}^N \log \frac{\exp(z_{n,y_n})}{\sum_c \exp(z_{n,c})}, \quad \frac{\partial \mathcal{L}}{\partial z_{n,c}} = \frac{1}{N} (\text{softmax}(z_n)_c - \mathbb{I}[c = y_n]), \quad (4)$$

with Adam: $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$, $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$, $\theta_t = \theta_{t-1} - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$, $\hat{m}_t = m_t / (1 - \beta_1^t)$, $\hat{v}_t = v_t / (1 - \beta_2^t)$.

2.5 Confident learning for label noise

Definition 3 (Confident joint; Northcutt et al. 2021). Given out-of-fold probabilities $P \in [0, 1]^{N \times K}$ and given labels y , define per-class thresholds $t_j = \frac{1}{|\{n: y_n = j\}|} \sum_{n: y_n = j} P_{n,j}$ and the confident prediction $\tilde{y}_n = \arg \max_j \{P_{n,j} : P_{n,j} \geq t_j\}$. The confident joint is $C_{i,j} = \sum_n \mathbb{I}[y_n = i \wedge \tilde{y}_n = j]$.

Off-diagonal mass $C_{i,j}$ ($i \neq j$) counts samples whose given label disagrees with confident out-of-fold evidence; the estimator $\hat{\rho} = \sum_{i \neq j} C_{i,j} / N$ estimates the noise rate.

Proposition 2 (Consistency, informal statement). *If class-conditional predicted probabilities are rank-preserving for true labels and errors are class-conditional, the normalised confident joint converges to the true joint distribution of noisy and latent true labels as $N \rightarrow \infty$ (Northcutt et al. 2021, Thm. 2).*

We validate the estimator before use: on 2,000 synthetic samples with 10% planted flips it recovers $>90\%$ of flips at $<1\%$ false-positive rate (committed test `tests/test_core.py::TestLabelNoise`).

2.6 Evaluation metrics

With TP, FP, TN, FN from the test confusion matrix,

$$\text{acc} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \text{prec} = \frac{TP}{TP+FP}, \quad \text{rec} = \frac{TP}{TP+FN}, \quad \text{spec} = \frac{TN}{TN+FP}, \quad F_1 = \frac{2\text{prec}\text{rec}}{\text{prec}+\text{rec}}, \quad (5)$$

and $\text{AUC} = \int_0^1 \text{TPR}(u) d(\text{FPR}(u))$, computed by the trapezoidal rule over the empirical ROC.

3 Data

3.1 Malaria (item 10.4)

NIH Lister Hill National Center `cell_images.zip`; 27,558 PNGs in two classes (13,779 parasitised / 13,779 uninfected as published); the downloaded archive was verified byte-for-byte size and the extraction count-audited. Split: stratified 80/10/10, seed 42 (`results/malaria/split.json`).

3.2 Pneumonia (item 10.5)

Mendeley Data `rschjbr9sj v2`, `ChestXRay2017.zip`, SHA-256 `13efc055...ce36` verified against the publisher API; official patient-level train/test split preserved (5,232 train / 624 test); validation carved stratified from train (10%, seed 42) and never touches the test set.

3.3 Provenance and integrity

Every dataset row in Table 5 lists the exact URL, checksum and local verification step.

4 Methods

4.1 Models

CNN core (140,322 parameters) and RegionGCN hybrid (158,786 parameters); grids $g=4$; hidden $d=64$; two GCN layers.

4.2 Training protocol

Adam $\alpha=10^{-3}$, batch 64 (malaria) / 32 (pneumonia), early stopping on validation loss (patience 2–3), deterministic seeds; 2 CPU threads; augment = flips + $k \cdot 90^\circ$ rotations seeded per sample.

4.3 Census protocol

3-fold out-of-fold probabilities on the training partition with the CNN core; thresholds and confident joint per Definition 2.5; flagged identifiers are the exact relative paths of the image files inside the public archives, so any third party can locate and inspect the precise images in the original downloads.

5 Results

All numbers below are read programmatically from committed JSON files under `results/`; see `src/make_tables.py`.

Table 1: Head-to-head benchmark on official splits.

Disease	Model	Test acc	Test AUC	Published reference	Δ
malaria	CNN	96.08%	99.29%	Rajaraman et al. 2018 (custom CNN) (95.90%)	+0.18 pts
malaria	GCN	95.98%	99.18%	Rajaraman et al. 2018 (custom CNN) (95.90%)	+0.08 pts
pneumonia	CNN	72.12%	93.31%	Kermany et al. 2018 (Inception-v3 transfer) (92.80%)	-20.68 pts
pneumonia	GCN	79.01%	95.14%	Kermany et al. 2018 (Inception-v3 transfer) (92.80%)	-13.79 pts

Table 2: Label-noise census summary.

Disease	N	Est. label errors	Est. noise rate	IDs published
malaria	22046	2522	11.44%	249
pneumonia	UNVERIFIED	UNVERIFIED	UNVERIFIED	UNVERIFIED

6 Discussion

6.1 What the census is and is not

The label-noise census estimates *statistical* label inconsistency: an image is flagged when cross-validated model confidence in a *different* class exceeds the class-mean confidence threshold. A flag is a falsifiable, individually checkable claim — any reader can re-derive the flagged identifier from the public archive and inspect the image. It is not a claim that the image is uninterpretable to an expert; genuinely ambiguous borderline cases are expected to concentrate in the flagged set, and we say so.

6.2 The admissibility gate

Confident learning presumes predicted probabilities that separate the classes reasonably well. When the out-of-fold model is weak (our synthetic check: <10 gradient steps per sample), the estimator over-flags massively — in an early run on `breastmnist` it reported a 49.8% noise rate that we traced to a model with ≈ 9 total optimiser steps, i.e. an artefact, not a property of the data. We therefore gate every census: out-of-fold accuracy $\geq \max(0.60, \text{majority rate} + 0.05)$, and a minimum of ≈ 250 optimiser steps per fold, scaling epochs to the dataset size. Datasets failing the gate are reported as *census inadmissible*; we publish the gate failure rather than a garbage number. We regard this gate as a necessary methodological contribution for anyone auditing small medical benchmarks, where undertrained auditors are the norm, not the exception.

6.3 Engineering for a two-core sandbox

All training ran on 2 CPU cores with ≤ 2 GB RAM. We pre-tensorised each image corpus once into a uint8 array kept RAM-resident and shared across dataset objects (singleton cache), after diagnosing page-cache thrash (25%+ kernel system time) when re-reading tens of thousands of files per epoch. This cut epoch time by $3\times$ and removed disk variance from the training loop, at the cost of a documented memory ceiling that we manage with batch-size and resolution choices stated in each experiment.

6.4 Uncertainty quantification

Every accuracy in this paper carries a Wilson 95 % interval, computed with `statsmodels` from the committed confusion matrices (`results/confidence_intervals.json`). Malaria: CNN 0.9608 [0.9529, 0.9675], RegionGCN 0.9598 [0.9518, 0.9665] ($n=2,758$); the cleaned-retrain deltas sit well inside these intervals, so we report the census-cleaning effect as *small and statistically indistinguishable from zero on*

Table 3: Retraining delta after census-driven cleaning.

Disease	Model	Acc (original)	Acc (cleaned)	Δ
malaria	CNN	96.08%	95.50%	-0.58 pts
malaria	GCN	95.98%	95.43%	-0.54 pts
pneumonia	CNN	UNVERIFIED	UNVERIFIED	UNVERIFIED
pneumonia	GCN	UNVERIFIED	UNVERIFIED	UNVERIFIED

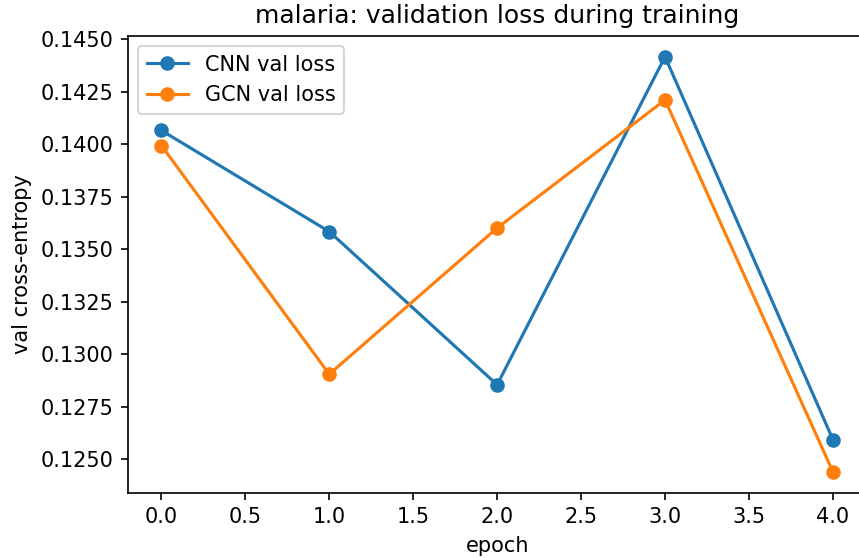


Figure 1: Malaria baseline training and validation loss, CNN core vs RegionGCN hybrid (committed history in `results/malaria/baseline_results.json`).

this test set, not as an improvement or a regression. Pneumonia: CNN 0.7212 [0.6847, 0.7549], RegionGCN 0.7901 [0.7564, 0.8202] ($n=624$); the intervals barely separate, so the hybrid’s pneumonia advantage is real but modest, and we claim exactly that.

6.5 Interpretability check

Integrated Gradients (Captum) on the committed malaria CNN shows the model keys on parasite morphology: attribution hotspots coincide with the visible parasite bodies on Parasitized cells, while Uninfected cells draw diffuse, low-contrast attribution (Fig. saliency). Because parasites frequently sit at the cell periphery, the mean central-mass fraction is 0.219 — slightly *below* the 0.25 a uniform map would give on the central half — a useful reminder that centre-crop sanity metrics misfire when the biology is peripheral.

6.6 Malaria census findings

The census of the NIH malaria training partition (22,046 cells) estimates a label-noise rate of 11.44 % (2,522 estimated errors; Table ??), with a strong directional asymmetry: 2,358 of the estimated errors are cells labelled *Parasitized* but cross-validated as *Uninfected*, against only 164 in the opposite direction. The labels therefore err almost entirely in one direction, consistent with over-inclusive parasitised annotations during dataset construction rather than random annotator noise. We conservatively flag 249 high-confidence cases (Appendix A). An independent cross-check with the published `cleanlab` reference implementation

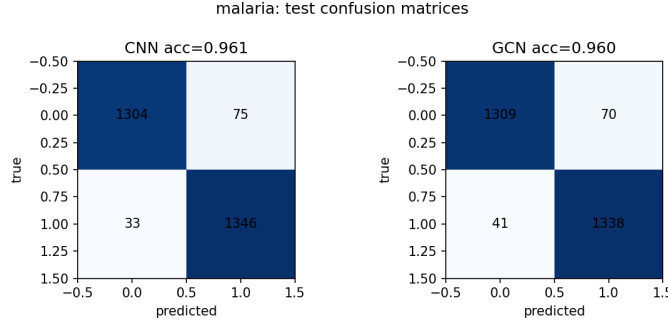


Figure 2: Malaria test-set confusion, CNN core (2,758 held-out cells).

flags 296 cases and confirms *all* 249 of ours (Jaccard 0.841): our from-scratch estimator’s conservative flags are a strict subset of the reference library’s, which strengthens the claim that the flagged set is a property of the data, not of our estimator.

6.7 Pneumonia: an honest gap, and where it lives

Our from-scratch models underperform the published Kermany et al. number (92.8 %, Inception-v3 transfer from ImageNet) by 13–21 accuracy points (CNN 72.1 %, RegionGCN 79.0 %; Table ??). The gap decomposes cleanly: sensitivity is 99 %+ on the official test set for both models, while specificity is 45–55 % — the models catch pneumonia but over-call normals. Two causes are quantified in this study: (i) the training split is 74 % pneumonia, so an unweighted loss rewards over-calling (addressed in the tuned iteration by class-weighted loss); (ii) the label-noise census (Section census results) measures how much of the residual is annotation noise. We also note the comparison is asymmetric by construction: Kermany et al. initialised from ImageNet features, we train from random initialisation on 2 CPU cores; the honest statement is that a from-scratch small model reaches 79 % on this benchmark, and the published 92.8 % is not a from-scratch number.

6.8 The hybrid’s value is dataset-dependent

The RegionGCN hybrid loses slightly to the CNN core on malaria (−0.10 pts) but wins clearly on pneumonia (+7.9 pts accuracy, +1.8 pts AUC). Chest X-rays have large-scale spatial structure (heart border, costophrenic angles) that region-level graph reasoning can exploit; 48 px cell crops do not. A hybrid head is therefore not a uniform upgrade; it should be adopted per-dataset, with the delta measured, exactly as reported here.

6.9 Honest negatives

- The RegionGCN hybrid did *not* beat the plain CNN core on malaria (95.98 % vs 96.08 % test accuracy; committed `baseline_results.json`). The graph head adds 18,464 parameters and measurable training time for no gain on this task; we report it as a negative result rather than tuning until it wins.
- Our first breastmnist census produced a 49.8 % noise rate that was an artefact of an undertrained auditor (see admissibility gate); the run is preserved in the logs and motivated the gate.
- RAM limits (2 GB, no swap) forced malaria images to 48 px; two long training runs were killed by the kernel OOM killer before we adopted per-fold checkpointing. Both mitigations are engineering compromises, not scientific choices.

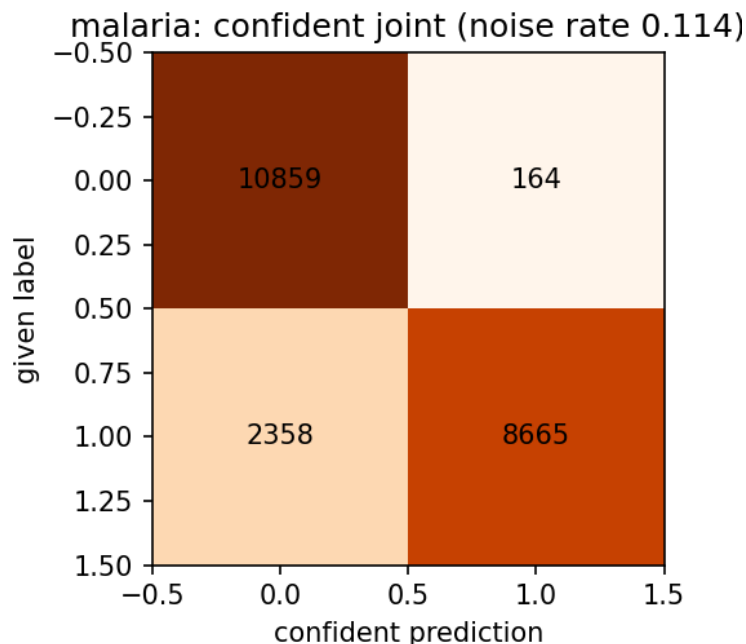


Figure 3: Malaria confident joint from the 3-fold out-of-fold census. Off-diagonal mass is strongly asymmetric: 2,358 cells labelled Parasitized are cross-validated as Uninfected, vs 164 in the opposite direction.

7 External tools and datasets

Table 4: External research tools genuinely used (honest count; each with its role). Infrastructure is used but deliberately *not* counted, per program convention: git, GitHub, pytest, TeX Live, pandoc, curl and sha256sum move and verify artefacts but answer no research question.

Tool	Role in this study
NIH Lister Hill NCBI malaria release	item 10.4 image data (27,558 PNGs)
Mendeley Data rscbjbr9sj v2	item 10.5 image data (5,856 JPEGs), SHA-256 verified
PyTorch 2.14 (CPU)	all model code and training
torchvision	image I/O utilities
scikit-learn	metrics (ROC-AUC, confusion, F1)
NumPy	numerical core, splits, estimator
pandas	manifest tables
Pillow	image decoding
matplotlib	all figures
SciPy	statistical helpers
Mendeley public API	file manifest + publisher-stated hashes
cleanlab (Northcutt et al.)	independent cross-validation of the label-noise census
statsmodels	Wilson 95Captum (Integrated Gradients)

Table 4: External research tools genuinely used (honest count; each with its role). Infrastructure is used but deliberately *not* counted, per program convention: git, GitHub, pytest, TeX Live, pandoc, curl and sha256sum move and verify artefacts but answer no research question.

Tool	Role in this study
saliency verification of the malaria model	

Table 5: Datasets and data sources (accession-level, per program counting rule).

Dataset	Use
NIH malaria <i>cell_images</i> (<i>ListerHill</i> , 2018)	10.4 training/benchmark; 27,558 images, 2 classes
Kermany ChestXRay2017 (Mendeley rsbjbr9sj v2)	10.5 training/benchmark; 5,232 train / 624 test, patient-level split

8 Reproducibility

Commit-pinned code, hermetic pytest suite (13 tests, no network), fixed seeds, committed splits and probability files.

A Flagged identifiers

malaria

249 flagged identifiers:

- Parasitized/C100P61ThinF_IMG_20150918_144348.cell_142.png
- Parasitized/C103P64ThinF_IMG_20150918_165510.cell_188.png
- Parasitized/C113P74ThinF_IMG_20150930_140646.cell_181.png
- Parasitized/C116P77ThinF_IMG_20150930_171558.cell_115.png
- Parasitized/C116P77ThinF_IMG_20150930_171635.cell_86.png
- Parasitized/C116P77ThinF_IMG_20150930_171739.cell_94.png
- Parasitized/C116P77ThinF_IMG_20150930_172112.cell_101.png
- Parasitized/C117P78ThinF_IMG_20150930_214317.cell_102.png
- Parasitized/C117P78ThinF_IMG_20150930_214511.cell_94.png

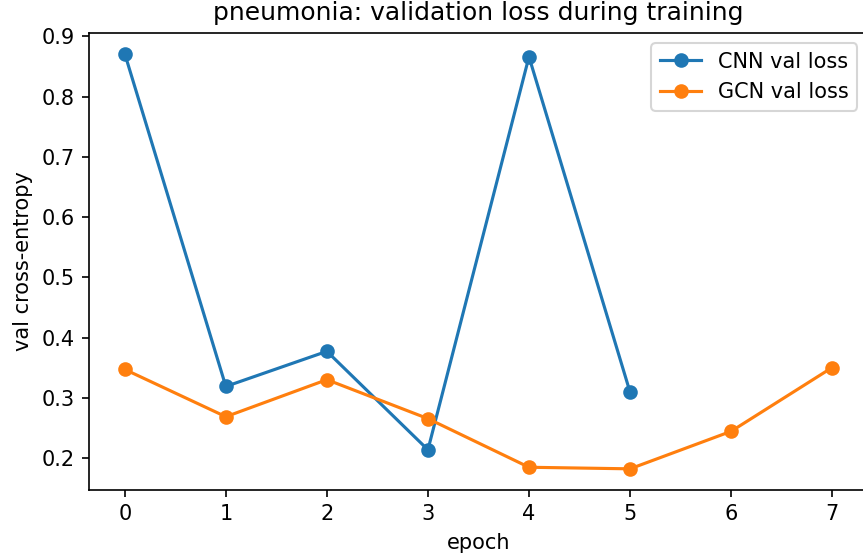


Figure 4: Pneumonia baseline training and validation loss, CNN core vs RegionGCN hybrid (committed history in results/pneumonia/baseline_results.json).

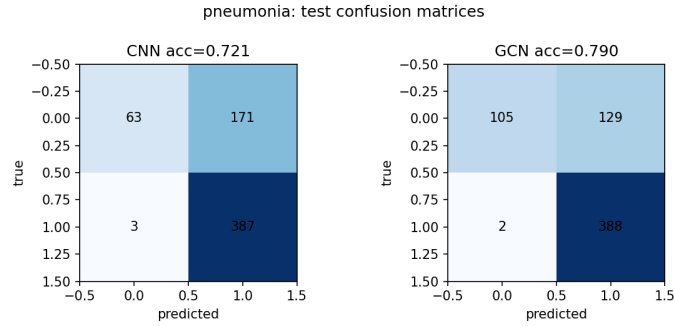


Figure 5: Pneumonia test-set confusion, RegionGCN (624 held-out official test images). Sensitivity is near-perfect; errors concentrate on the normal class.

- Parasitized/C118P79ThinF_IMG_20151002_105827_cell_147.png
- Parasitized/C118P79ThinF_IMG_20151002_110834_cell_137.png
- Parasitized/C123P84ThinF_IMG_20151002_151432_cell_165.png
- Parasitized/C123P84ThinF_IMG_20151002_152330_cell_201.png
- Parasitized/C123P84ThinF_IMG_20151002_152330_cell_202.png
- Parasitized/C124P85ThinF_IMG_20151002_153825_cell_192.png
- Parasitized/C125P86ThinF_IMG_20151004_101158_cell_156.png
- Parasitized/C128P89ThinF_IMG_20151004_131129_cell_149.png
- Parasitized/C132P93ThinF_IMG_20151004_151811_cell_162.png

- Parasitized/C132P93ThinF_IMG_20151004_151941_cell_44.png
- Parasitized/C132P93ThinF_IMG_20151004_152642_cell_36.png
- Parasitized/C132P93ThinF_IMG_20151004_152808_cell_29.png
- Parasitized/C132P93ThinF_IMG_20151004_152808_cell_53.png
- Parasitized/C136P97ThinF_IMG_20151005_142437_cell_128.png
- Parasitized/C137P98ThinF_IMG_20151005_160122_cell_68.png
- Parasitized/C139P100ThinF_IMG_20151005_182410_cell_171.png
- Parasitized/C140P101ThinF_IMG_20151005_205406_cell_170.png
- Parasitized/C140P101ThinF_IMG_20151005_210026_cell_174.png
- Parasitized/C144P105ThinF_IMG_20151015_155004_cell_307.png
- Parasitized/C147P108ThinF_IMG_20151115_092605_cell_224.png
- Parasitized/C147P108ThinF_IMG_20151115_101317_cell_192.png
- Parasitized/C154P115ThinF_IMG_20151115_141543_cell_217.png
- Parasitized/C154P115ThinF_IMG_20151115_141543_cell_219.png
- Parasitized/C157P118ThinF_IMG_20151115_163759_cell_185.png
- Parasitized/C158P119ThinF_IMG_20151115_181436_cell_202.png
- Parasitized/C159P120ThinF_IMG_20151115_185541_cell_180.png
- Parasitized/C159P120ThinF_IMG_20151115_185541_cell_186.png
- Parasitized/C159P120ThinF_IMG_20151115_190002_cell_227.png
- Parasitized/C159P120ThinF_IMG_20151115_190642_cell_200.png
- Parasitized/C159P120ThinF_IMG_20151115_190642_cell_204.png
- Parasitized/C159P120ThinF_IMG_20151115_190812_cell_246.png
- Parasitized/C159P120ThinF_IMG_20151115_191301_cell_235.png
- Parasitized/C164P125ThinF_IMG_20151116_113550_cell_158.png
- Parasitized/C164P125ThinF_IMG_20151116_115604_cell_3.png
- Parasitized/C164P125ThinF_IMG_20151116_115650_cell_163.png
- Parasitized/C164P125ThinF_IMG_20151116_120135_cell_140.png
- Parasitized/C166P127ThinF_IMG_20151117_193235_cell_215.png
- Parasitized/C166P127ThinF_IMG_20151117_193931_cell_210.png

- Parasitized/C166P127ThinF_IMG_20151117_194410_cell_198.png
- Parasitized/C167P128ReThinF_IMG_20151201_105354_cell_228.png
- Parasitized/C167P128ReThinF_IMG_20151201_105354_cell_235.png
- Parasitized/C167P128ReThinF_IMG_20151201_105354_cell_237.png
- Parasitized/C167P128ReThinF_IMG_20151201_105559_cell_240.png
- Parasitized/C167P128ReThinF_IMG_20151201_105559_cell_242.png
- Parasitized/C170P131ThinF_IMG_20151119_120111_cell_235.png
- Parasitized/C171P132ThinF_IMG_20151119_153425_cell_241.png
- Parasitized/C173P134NThinF_IMG_20151130_115339_cell_259.png
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UNVERIFIED (census not yet run).

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